

Midterm Exam 1
CENG 113, SP26
Dr. Drews

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Question 1: Estimate K_0 at 500 K for a liquid-phase equilibrium reaction with $\Delta G_{T_R}^\circ = 3 \text{ kJ mol}^{-1}$, $\Delta H_{T_R}^\circ = 10 \text{ kJ mol}^{-1}$, and $\Delta C_P = 0$.

$$K_0 = \exp(-\Delta G/RT) = \exp(-3000/(8.314 \cdot 298)) = 0.298$$

Question 2: Estimate K_1 given the information in the previous problem.

$$K_1 = \exp(\Delta H/R \cdot (1/T_R - 1/T)) = \exp(10000/8.314 \cdot (1/298 - 1/500)) = 5.11$$

Question 3: What's the elementary rate expression for the reaction $A \rightleftharpoons B + C$?

- (A) $r = kC_A$
- (B) $r = kC_B C_C$
- ☒ (C) $r = k \left(C_A - \frac{C_B C_C}{K_C} \right)$
- (D) $r = k \left(C_B C_C - \frac{C_A}{K_C} \right)$
- (E) none of the above

Question 4: Volumetric flow rate is always constant for ideal gases if the system is isobaric and adiabatic.

- (A) true
- ☒ (B) false
- (C) not enough information to decide

Question 5: Enthalpy of reaction is maximized at equilibrium.

- (A) true
- ☒ (B) false
- (C) not enough information to decide

Question 6: Select the best activity for a pure solid.

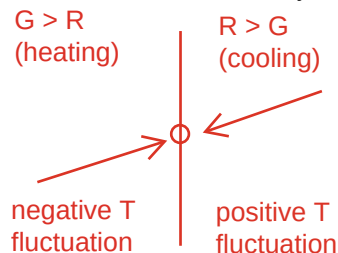
- (A) P_i/P°
- (B) M_i/M_i°
- (C) x_i
- ☒ (D) 1
- (E) none of the above

Question 7: The equilibrium constant K_a never has units.

- ☒ (A) true
- (B) false
- (C) not enough information to decide

Question 8: If $R > G$ above a steady-state operating temperature and $G > R$ below, this steady state is best classified as which of the following?

- ☒ (A) stable
- ☐ (B) unstable
- ☐ (C) metastable
- ☐ (D) not enough information to decide



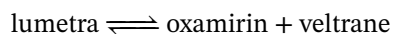
Question 9: If the reaction $A \rightleftharpoons B$ is rewritten as $2A \rightleftharpoons 2B$ then $\Delta G_{T_R}^\circ$ will double.

- ☒ (A) true
- ☐ (B) false
- ☐ (C) not enough information to decide

Question 10: Estimate the CSTR reactor volume (in m^3) needed to convert 70% of A according to the liquid-phase, elementary reaction $A \longrightarrow \text{products}$ ($k = 3 \text{ min}^{-1}$) assuming a feed of pure A at $0.2 \text{ m}^3 \text{ min}^{-1}$.

$$V = nA_0 X_a / (-r_A) = nA_0 X_a / (k C_{A0} (1 - X_a)) = (v_0 / k) X_a / (1 - X_a) = (0.2 / 3) * 0.7 / (1 - 0.7) = 0.155 \text{ m}^3$$

Question 11: For the gas-phase equilibrium reaction



determine the pressure (in kPa) to produce 0.2 mol s^{-1} oxamirin from a 1 mol s^{-1} feed of pure lumetra at 575 K.

Sp	nu	ni0	ni
A	-1	1	0.8
B	1	0	0.2
C	1	0	0.2

species	liquid phase		gas phase	
	$\Delta H_{f,i}^\circ$	$\Delta G_{f,i}^\circ$	$\Delta H_{f,i}^\circ$	$\Delta G_{f,i}^\circ$
lumetra	-277	-174	-235	-167
oxamirin	-173	-9	-146	-8
veltrane	-285	-237	-241	-228
zenthyl	-156	26	-123	31

$$dH = -152 \quad dG = -69$$

$$K_0 = \exp(-dG/RT) = \exp(69000/8.314/298) = 1.244e12$$

$$K_1 = \exp(dH/R*(1/TR - 1/T)) = \exp(-152000/8.314*(1/298 - 1/575)) = 1.46e-13$$

$$K_a = K_{sc} = K_0 * K_1 = (1.244e12) * (1.46e-13) = 0.182$$

$$\text{prod}(a_i^{\nu_i}) = a_C * a_B / a_A = y_C * y_B / y_A * (P/P_0) = n_C * n_B / (n_A * n_T) * (P/P_0)$$

$$= (0.2 * 0.2) / (0.8 * 1.2) * (P/P_0)$$

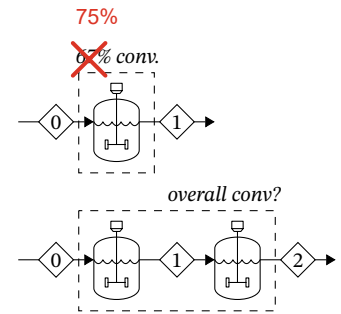
$$= 0.0416 * P/P_0$$

$$K_a = 0.182 = 0.0416 * P/P_0$$

$$P = 4.36 * P_0 = 440.6 \text{ kPa}$$

(can vary about 430 - 460 depending on digits retained throughout)

Question 12: The elementary, liquid-phase transmodulysis reaction $A \rightleftharpoons B$ has equilibrium constant $K_C = 4$. Compounds A and B are miscible and of equal density. A feed of pure A achieves 75% conversion in a single isothermal CSTR. Determine the *overall* conversion of A if the product from the first reactor is sent directly to a second reactor with the same operating conditions as the first (temperature, volume, etc.).



In both cases $r = k^*(C_A - C_B/K_C) = k^*C_{A0}^*(1-X_A - (\theta_B + X_A)/K_C)$

Assume a basis of 4 mol/s A fed (not required but helps with math)

FIRST REACTOR

Sp	nu	ni0	ni
A	-1	4	1
B	1	0	3

$$V = n_{A0}X_A/(-r_A) = n_{A0}X_A/(k^*C_{A0}^*(1-X_A - (0+X_A)/K_C))$$

from which

$$\tau k = X_A/(1-X_A-X_A/K_C) = 12 = \text{constant (same temp, same } v_0, \text{ latter b/c same density)}$$

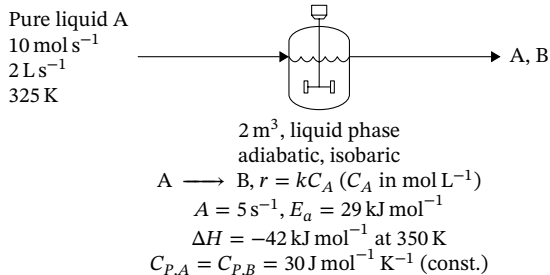
SECOND REACTOR

Sp	nu	ni0	ni
A	-1	1	$1^*(1-X_A)$
B	1	3	$1^*(3+X_A)$

$$\tau k = X_A/(1-X_A-(3+X_A)/K_C) = 12 \quad \text{from which } X_A = 0.1875$$

$$\text{Overall conversion: } X_{A,ov} = 1 - n_{A2}/n_{A0} = 1 - 1^*(1-0.1875)/4 = 0.797 = 79.7\%$$

Question 13: On the back of your answer sheet write a MATLAB script which uses fsolve to determine the outlet molar flow rate of all species and temperature for the adiabatic reactor shown below. Use the general layout of a script with a local function described in class (other approaches, such as symbolic math or anonymous functions, will not receive credit).



```

1 % Y = [nA, nB, T] in [mol/s, mol/s, K]
2 Y = fsolve(@myRFF, [5 5 375]); % guess can be anything
3
4 function out = myRFF(Y)
5 nA = Y(1); nB = Y(2); T = Y(3); % mol/s or K
6 v = 2; % L/s, = v0 since liq
7 V = 2000; % L
8 Ca = nA/v; % mol/L
9 r = 5*exp(29000/(8.314*325))*Ca; % mol/L-s
10 dH = -42e3; % J/mol (const since dCp = 0)
11
12 out(1,1) = 10 - r*V - nA; % mb on A
13 out(1,2) = 0 + r*V - nB; % mb on B
14 out(1,3) = 10*30*(T-325) + r*V*dH; % eb with Q = 0
15 end

```